

Therapeutic Activities

For Residents with Autism & Developmental Disabilities

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The Problem

What we observed at Flicker and why it matters

Lack of Personalization

Therapeutic activities are not adapted to individual resident abilities, needs, or current emotional state.

Low Engagement & Wellbeing

Without matched activities, residents show low participation, increased stress, restlessness, and reduced quality of interaction.



Staff Cognitive Burden

Staff especially new or substitute caregivers carry the burden of identifying, selecting, and justifying the right activity in real time, every shift.

This system addresses all three barriers — through structure, evidence, and autonomous decision-making.

Context

Who this system is for and when it is used

WHO

- Residents of Flicker with autism & developmental disabilities
- Support staff full-time, new, and substitute caregivers

WHERE

- Common rooms, quiet rooms, and activity spaces

WHEN

- Weekly — scheduled group sessions (e.g. Monday and Friday)
- Need-based — triggered dynamically by resident state or logbook input
- During periods of inactivity or disengagement
- During situations of stress or emotional overwhelm



Pepper works across all these settings — adapting to each resident's needs in real time.

Key Insight — Why These Activities?

Every activity is grounded in evidence from structured care research

Research in structured care environments indicates:

- Personalized activities significantly improve engagement and emotional well-being (Kolanowski et al., 2018)
- Structured, simple routines enable staff to act consistently and with confidence (Kitwood, 1997)
- Sensory stimulation reduces agitation especially in low-cognitive states (Ayres, 1972)
- Music activates motor, emotional, and cognitive brain networks (Thaut & Hoemberg, 2014)
- Breathing and movement reduce physiological stress responses (Porges, 1994)

Selected therapeutic approaches:



Sensory
stimulation



Chair yoga



Conversation
& storytelling



Music
therapy



Breathing &
meditation

Proposed Solution

A simple, adaptable system that autonomously delivers the right activity to the right person

The system is designed to:

- Personalize activities using individual resident profiles via the Logbook and facial recognition
- Reduce staff cognitive load staff no longer need to decide which activity is appropriate in the moment
- Support new or substitute staff who may be unfamiliar with individual resident needs and preferences
- Deliver multi-sensory activities
- Be easy to implement with minimal preparation in any care setting



How It Works

A step-by-step view of how Pepper and staff work together

1

Staff identifies a moment of disengagement or stress

Observed directly or triggered by a logbook entry from a previous shift.

2

Resident profile is retrieved

Via facial recognition or logbook Pepper identifies who the resident is and pulls their profile.

3

The system autonomously selects a suitable activity

Based on cognitive level, mobility, preferences, and current logbook state. Staff does not choose.

4

Step-by-step guidance is followed

Pepper leads the session. The activity is structured, evidence-based, and predictable.

5

Resident engages through various therapeutic activities

Activities are designed to meet the resident where they are.

6

Staff observes responses and adapts if needed

Pepper guides but staff remain present and supportive throughout.

7

Outcomes are recorded in the logbook

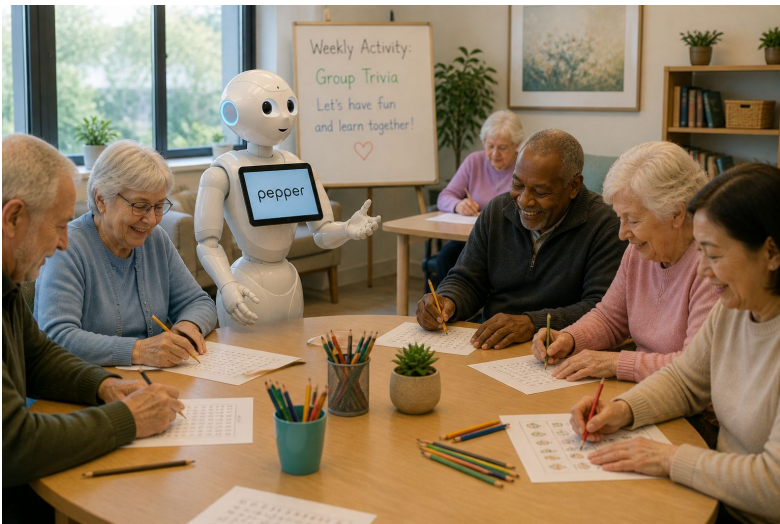
Mood, engagement, and notes logged for continuity of care across shifts.

Two Activity Models — One Unified Purpose

Structured therapeutic care for every resident, every shift

GROUP MODE

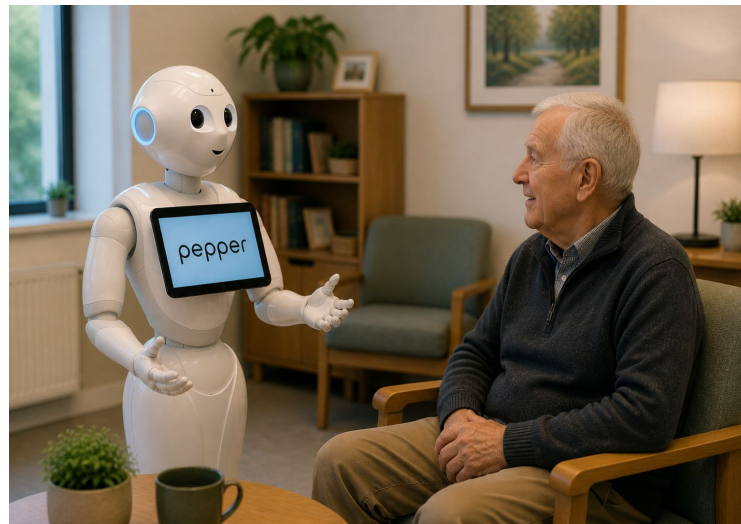
Weekly Activities



AND

PERSONAL MODE

Need-Based Activities



Weekly Activities (4 Groups)

Choose activities based on the resident's cognitive / communication level and mobility level

High Cognitive + High Mobility

Qualifications

- Resident is physically active and mobile
- Able to follow instructions and engage cognitively
- Good communication and interaction skills

Activity

Music Choice + Active Movement

(Neurologic Music Therapy — Thaut & Hoemberg, 2014)

High Cognitive + Low Mobility

Qualifications

- Limited physical movement
- Able to understand, communicate, and engage mentally
- Prefers seated or stationary engagement

Activity

Music Therapy — Seated Sing-Along / Karaoke

(Neurologic Music Therapy — Thaut & Hoemberg, 2014)

Low Cognitive + High Mobility

Qualifications

- Physically active and energetic
- Needs simple instructions and visual guidance
- Limited verbal or cognitive engagement

Activity

Chair Yoga — Gentle Movement Sequence

(Biopsychosocial Model — Engel, 1977)

Low Cognitive + Low Mobility

Qualifications

- Limited movement and communication
- Needs calm, simple, sensory input
- Easily overstimulated or fatigued

Activity

Chair Yoga — Seated Stretch & Hand Movements

(Biopsychosocial Model — Engel, 1977)

This framework helps staff choose the most appropriate weekly activity based on each resident's current abilities.

Why These Quadrants?

The framework is grounded in two dimensions widely used in care and therapy

The Framework is Based on:



Cognitive / Communication Ability



Mobility Level

These dimensions help care teams to:

- Adapt activities to individual abilities
- Improve resident engagement and participation
- Reduce overstimulation or frustration
- Ensure every resident is included

1

High Cognitive + High Mobility

Active engagement and cognitive challenge improves mood and motivation

Flow Theory — Csikszentmihalyi

2

High Cognitive + Low Mobility

Cognitive stimulation without physical strain preserves engagement and reduces frustration

Person-Centred Care — Kitwood (1997)

3

Low Cognitive + High Mobility

Simple, repetitive, visual activities promote participation and reduce overstimulation

Sensory Integration Theory — Ayres (1972)

4

Low Cognitive + Low Mobility

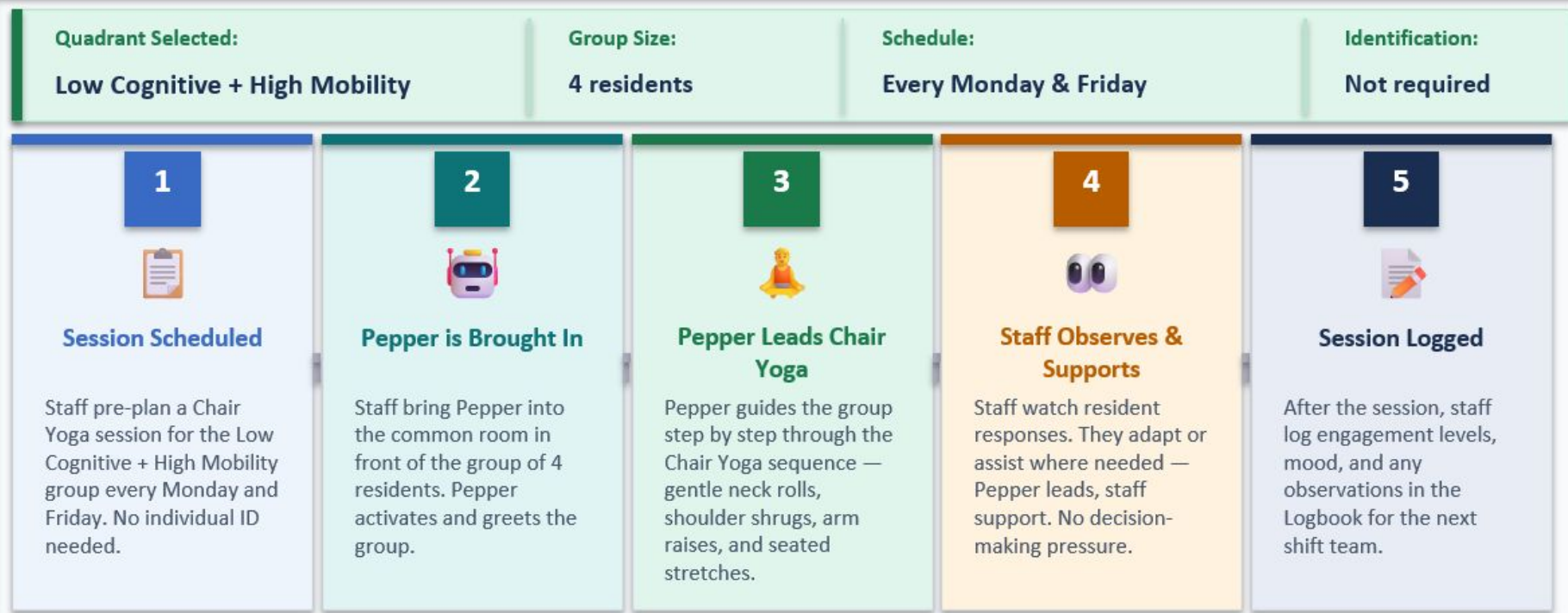
Sensory stimulation reduces anxiety and agitation in low-ability states

Biopsychosocial Model — Engel (1977)

These four quadrants ensure every resident receives an activity matched to their current ability — no one is left out.

Weekly Activity — A Real Example

How Pepper leads a scheduled group session step by step



Based on: Biopsychosocial Model (Engel, 1977) · Simple, repetitive movement reduces overstimulation in low-cognitive states

This same flow repeats every scheduled session — consistent, structured, and staff-friendly. Pepper does the leading, staff do the caring.

Why Chair Yoga?



Sensory Regulation

Slow, predictable movements and guided breathwork activate the parasympathetic nervous system — reducing sensory overload and calming anxiety in autistic individuals.



Motor Coordination & Balance

Chair yoga safely improves gait, balance, and functional mobility in high-mobility adults — all participants in clinical studies showed improved Tinetti balance scores.



Emotional Self-Regulation

Structured sequences with clear, simple instructions provide predictability — a key support for autism. Research shows reductions in anxiety, agitation, and hyperactivity.



Safe Group Engagement

The seated format removes fall risk, making participation low-barrier. Group sessions foster social presence without overwhelming cognitive demands for low-cognitive participants.

Weekly Activity

Today: Wednesday, 21 May 2025

10:15 AM

Weekly Activities

Need-Based Activities

Residents

Activity Library

Progress

Settings

Hello! I'm Pepper.

I'm here to guide our activities together.

Chair Yoga Session

Weekly Activity

In Progress

Side Stretch

Raise your right arm over your head and gently lean to the left. Hold... and return.

15 seconds

Current Step

1 Warm Up 2 min

>

2 Side Stretch 2 min

>

3 Arm Raise 2 min

>

4 Torso Twist 2 min

>

5 Cool Down 2 min

Tip from Pepper

Breathe in through your nose... and out through your mouth.

Activity Overview

Chair Yoga

30 minutes

Group Activity

All Ability Levels

Participants (6)

David Participating

Maria Participating

John Participating

Aisha Participating

Peter Needs Support

Linda Participating

View All Participants

Session Progress

60%

18 / 30 min completed

☐ End Activity

Weekly Activity — Another Example

How Pepper presents music activities to a scheduled group session

Quadrant Selected:

High Cognitive + High Mobility

Group Size:

4–6 residents

Schedule:

Every Monday & Friday

Identification:

Not required

Why These Activities?

Cognitive Stimulation

Singing familiar songs and listening to music activate memory, attention, and language centres simultaneously — sustaining function in cognitively intact older adults.

Motor Engagement

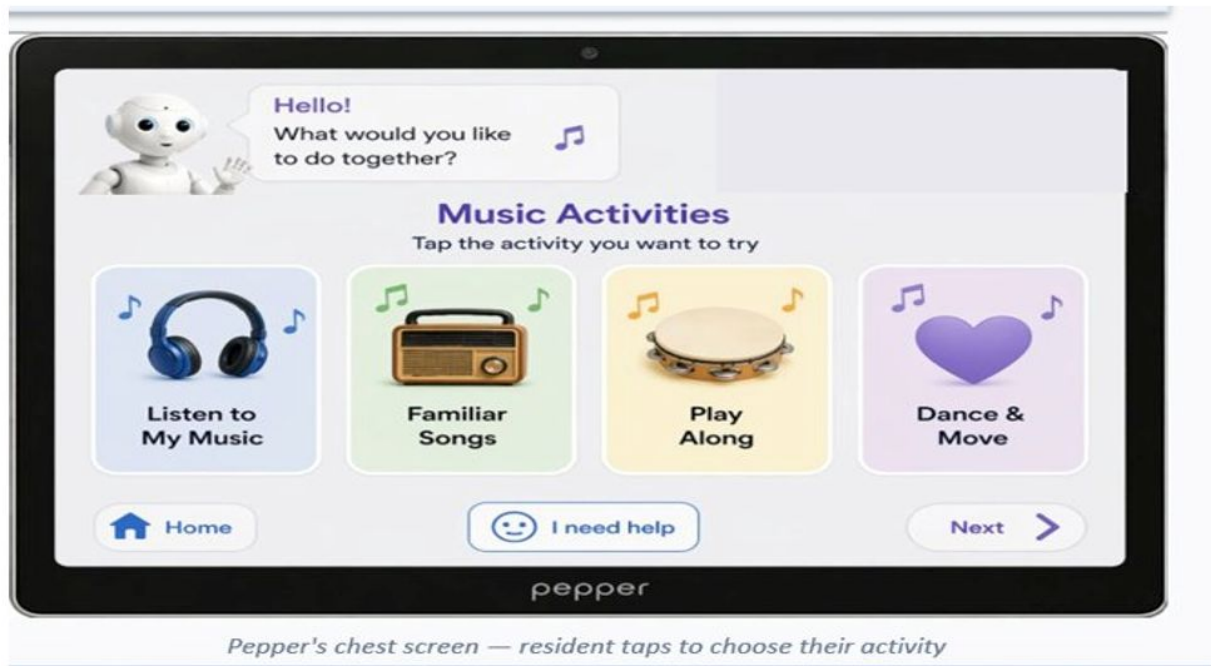
Rhythmic play-along activities use auditory rhythm entrainment to coordinate movement, improving gait timing and fine motor control for high-mobility residents.

Mood & Emotional Balance

Music triggers dopamine release, reducing anxiety and agitation — helping high-ability residents stay calm, motivated, and emotionally regulated during group sessions.

Social Bonding

Shared musical experiences foster group cohesion. Live, participatory music elicits stronger socio-emotional engagement than passive listening alone.



Meet David.

A real example of how Pepper personalizes care.



David

Age:	72
Cognitive Level:	High
Mobility Level:	Low
Preferred Time:	Morning
Favourite Activities:	Music · Relaxation

What is Need-Based?

Unlike weekly sessions, need-based activities are triggered dynamically — based on who the resident is and what they need right now.



Facial Recognition Trigger

Resident stands in front of Pepper. Pepper scans their face, identifies them, retrieves their profile.



Logbook Trigger

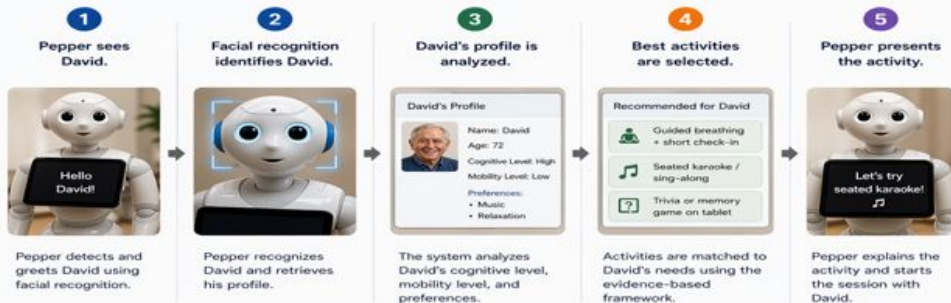
Staff note in the Logbook that a resident had a bad night or is agitated. Next shift sees this and Pepper adapts.

Key difference from Weekly: Staff do NOT choose the activity. Pepper selects it autonomously based on David's profile and current state.

Need-Based — Step 1: Facial Recognition

David walks in front of Pepper → Pepper identifies him → retrieves his profile → system selects an activity

HOW IT WORKS WITH PEPPER



Personalized. Evidence-based. Engaging.
Every activity is chosen for the person, not for the group.

FACIAL RECOGNITION IN ACTION



Recognition Successful



Name: David
Age: 72
Cognitive Level: High
Mobility Level: Low

Last Activity:
Seated Karaoke
Preferred Time: Morning

Activity Insight
David enjoys music and responds well to calm, seated activities.



Pepper uses facial recognition to deliver the right activity to the right person every time.

WHAT IT LOOKS LIKE IN THE PROTOTYPE (STAFF / TABLET VIEW)



CUSTOMIZABLE & CONTINUOUSLY LEARNING



Staff can adjust, add, or skip activities based on David's current mood and needs.



Pepper learns from each interaction to improve future recommendations.



All data is stored securely and used only to support personalized care.

REFERENCES

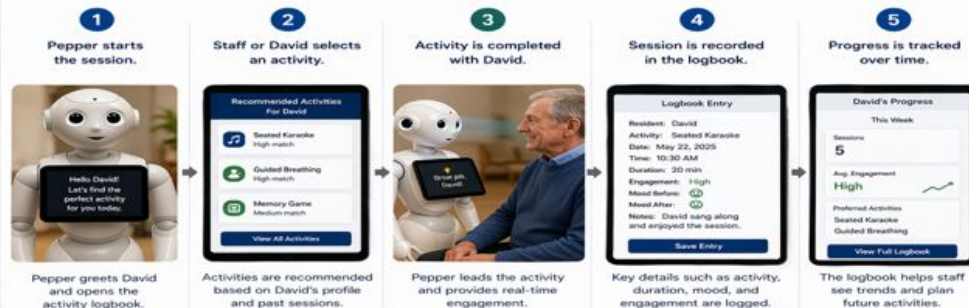
1. Kolanowski et al., 2018 2. Gitlin et al., 2008 3. Lawton, 1975 4. Zeisel et al., 2013 5. Kitwood, 1997 6. O'Shea et al., 2014 7. O'hang et al., 2015 8. Clare et al., 2017 9. Livingston et al., 2014 10. McCormick et al., 2016 11. Ploeg et al., 2008 12. Buettner et al., 2016 13. Seitz et al., 2017

Pepper autonomously identifies David and selects the most appropriate activity — no staff decision required.

Need-Based — Step 2: The Logbook

Staff log David's session → mood, engagement, and notes recorded → next shift can see exactly what worked

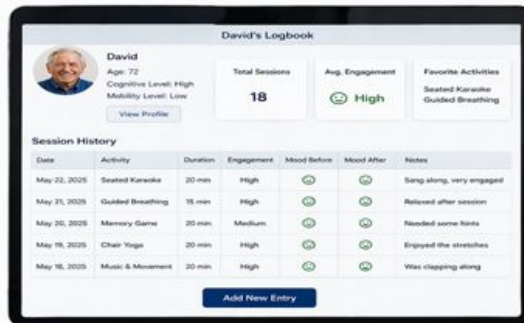
HOW IT WORKS WITH PEPPER (USING THE LOGBOOK)



Consistent. Personalized. Evidence-based.

Every session is logged to support person-centered care and improve outcomes.

THE LOGBOOK: A COMPLETE RECORD



What the Logbook Captures

- Activity Details**
What activity was done and for how long.
- Engagement Level**
How engaged David was during the session.
- Mood Tracking**
David's mood before and after the activity.
- Notes**
Important observations from staff.
- Trends Over Time**
Helps staff understand what works best.



Better insights. Better care.

The logbook turns every interaction into valuable insights for David's well-being.

WHAT IT LOOKS LIKE IN THE PROTOTYPE (STAFF / TABLET VIEW)



WHY THE LOGBOOK MATTERS

- Continuity of Care**
Ensures every staff member is informed about David's preferences and progress.
- Person-Centered**
Activities are tailored based on what David enjoys and responds to.
- Better Outcomes**
Tracking engagement and mood helps improve quality of life.
- Data for Decisions**
Helps teams make informed choices and show impact.



All logbook data is stored securely and used only to support personalized care.

The Logbook eliminates reliance on staff memory — every shift handover is clear, structured, and data-driven.

Need-Based Activity Flow — How Pepper Serves David

Facial Recognition → Profile Analysis → Activity → Logbook Record



STEP 6 — Session Recorded in the Logbook

Pepper's Memory for David

What is the Logbook?

The Logbook is a structured digital record maintained by support staff after every session with David. It is the system's memory ensuring every shift, every carer, knows exactly where David is.

It captures:

- Resident mood before and after each session
- Level of engagement — high, medium or low
- Notable events — poor sleep, agitation, refusal
- Staff observations and notes per session
- Trends over time — what works, what doesn't

Is the activity working?

Tracks whether David engaged well or showed disinterest — so activities can be updated or swapped.

Is this right for David?

Over time the logbook reveals what truly works for David — so each session gets better and more personalised.

Mood Before & After

Records David's emotional state before and after the session — helping staff spot patterns and triggers.

Continuity Across Shifts

Every carer on any shift sees exactly what happened — no reliance on memory or handover word-of-mouth.

The Logbook eliminates reliance on staff memory — every shift handover is clear, structured, and data-driven.

Activity 1 — Sensory & Touch Activities

Need-Based Activity · Triggered by David's current mood and state

Triggered when David is feeling:

 Agitated

 Angry

 Overwhelmed

 Withdrawn

Why We Chose This Activity

- Sensory overload can cause agitation and withdrawal in autistic individuals.
- Gentle tactile input activates the calming parasympathetic nervous system.
- This provides immediate relief without requiring cognitive effort.

Impact:

- ✓ Restore calm
- ✓ Builds emotional regulation skills over time
- ✓ Improves mood and sense of comfort

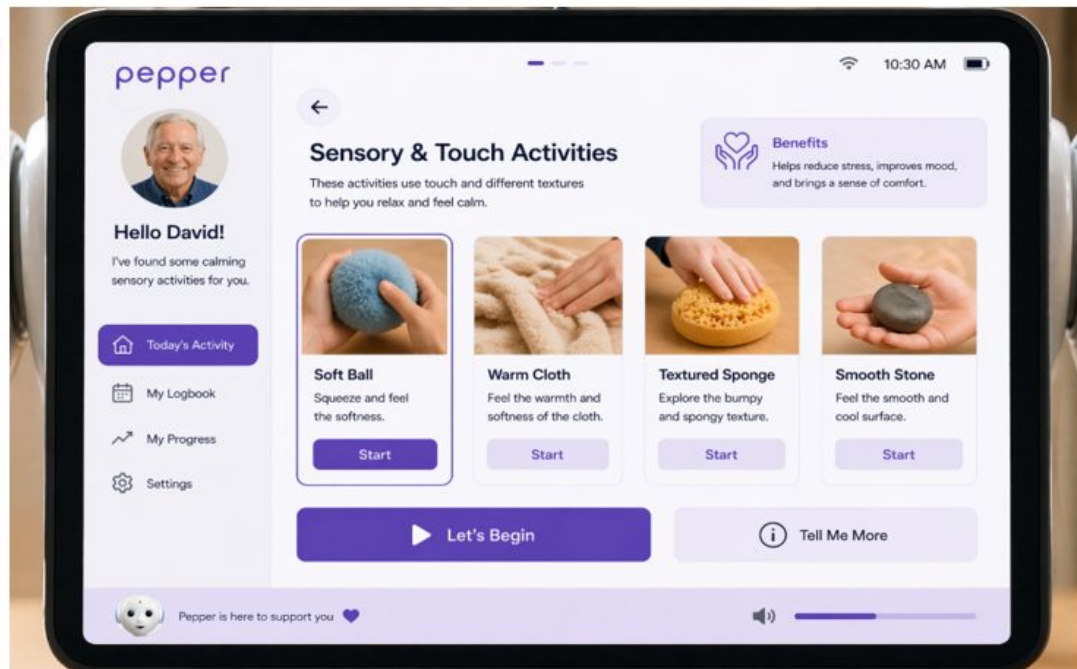
Research Behind This Activity

Sensory stimulation lowers cortisol and physiological related issues in autistic individuals

Ayres, Sensory Integration Theory (1972) · Grandin & Panek (2013)

Tactile input activates the parasympathetic nervous system, shifting the body from stress to calm

Porges, Polyvagal Theory (1994) · Lane et al. (2019)



Pepper's screen — displayed to David during the session

Sensory activities help David self-regulate — reducing agitation through controlled tactile stimulation.

Activity 2 — Breathing Exercise with Pepper

Need-Based Activity · Triggered by David's current mood and state

Triggered when David is feeling:



Angry



Anxious



Stressed



Tense

Why We Chose This Activity

- Triggered when David feels angry, anxious, stressed, or tense.
- Slow breathing activates the vagus nerve and reduces stress response.
- Helps lower anxiety and prevent emotional meltdowns.

Impact:

- ✓ Reduces anger and anxiety
- ✓ Reduces the frequency and intensity of emotional meltdowns

Research Behind This Activity

Slow diaphragmatic breathing activates the vagus nerve and directly counters the stress response

Porges, Polyvagal Theory (1994) · Zaccaro et al., Frontiers in Human Neuroscience (2018)

Guided breathing reduces anxiety and meltdown frequency in autistic individuals

Gerburg & Brown, Journal of Child & Adolescent Psychopharmacology (2015)

 Interaction with Pepper

Resident: DavidTime: 14:30



Pepper

Hi David, I see you might need a relaxing moment. Would you like to do a short activity together?

14:30

David

(Looks at Pepper and nods slightly)

14:31

Pepper

Great! Let's try a simple breathing exercise. Follow me.

"Breathe in... slowly... and out..."

14:31

David

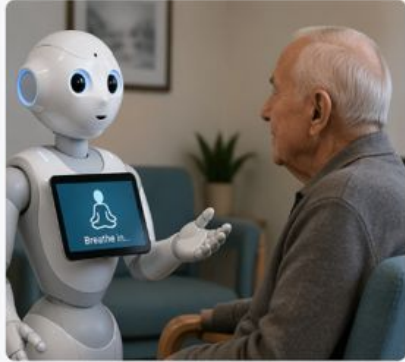
(Follows breathing, body relaxes)

14:32

Pepper

You're doing very well. Would you like to continue, or try some calming music?

14:32



Suggested Activities

 Continue Breathing Exercise>

 Try Calming Music>

Pepper's screen — displayed to David during the session

Breathing exercises help David move from anger or anxiety to calm — quickly and without medication.

Activity 3 — Conversation with Pepper

Need-Based Activity · Triggered by David's current mood and state

Triggered when David is feeling:

😞 Sad

😞 Low mood

😞 Lonely

😞 Disengaged

Why We Chose This Activity

- Triggered when David feels sad, lonely, low mood, or disengaged.
- Structured conversation supports emotional expression and connection.
- Improves mood and increases verbal engagement.

Impact:

- ✓ Lifts mood
- ✓ Encourages verbal expression and emotional processing
- ✓ Sense of connection and being heard

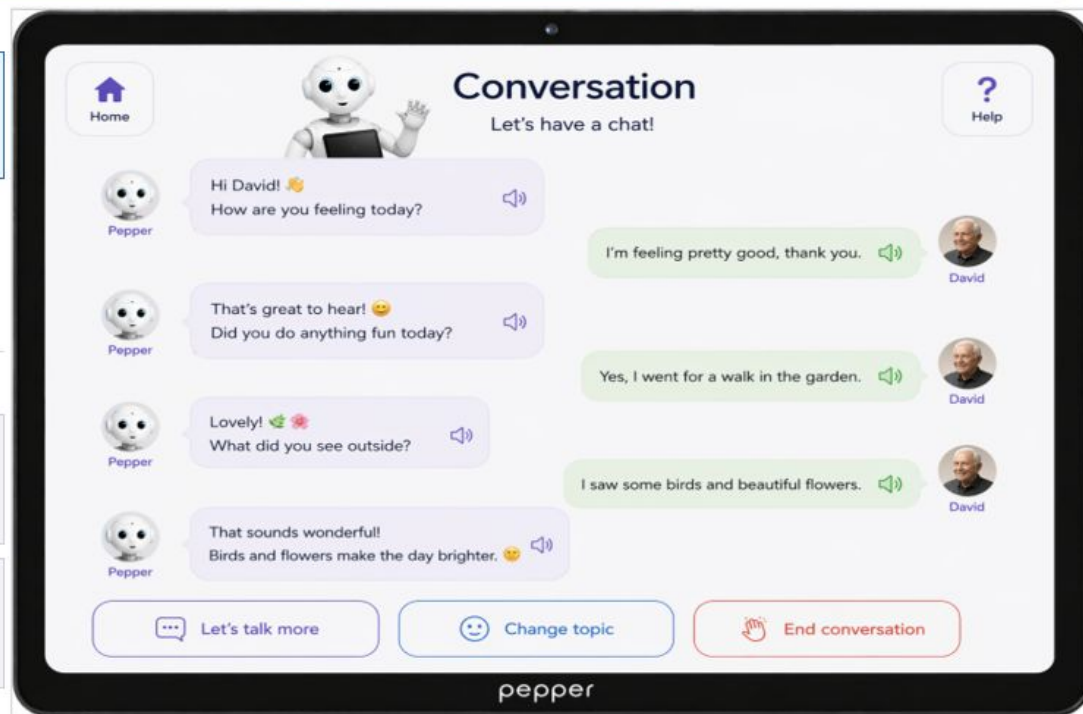
Research Behind This Activity

Social interaction and conversation stimulate dopamine release, improving mood in autistic individuals

Bandura, Social Learning Theory (1977) · Chevallier et al., TICS (2012)

Robot-assisted conversation increases verbal engagement and emotional expression in autistic adults

Wainer et al., Journal of Autism & Developmental Disorders (2014)



Pepper's screen — displayed to David during the session

Conversation with Pepper helps David feel heard and connected — lifting his mood when he is sad or withdrawn.

Impact & Strategic Value

What this system delivers — for residents and for staff

For Residents

- Increased participation and engagement
- Reduced stress, agitation, and distress
- Improved mood and emotional regulation
- Consistent, predictable therapeutic routines
- Activities matched to individual needs and preferences

For Staff

- Removes burden of real-time activity selection
- Reduces cognitive load especially for new or substitute staff
- Shift handover is clear, structured, and data-driven
- Consistent care delivery regardless of staff experience
- Staff focuses on support not decision-making

Next Steps

From prototype to implementation — a phased roadmap

Phase 1

Pilot Testing

- Deploy prototype in a live care setting at Flicker
- Observe resident and staff interactions with the system
- Identify usability gaps and activity suitability

Phase 2

Feedback & Refinement

- Collect structured feedback from support staff and caregivers
- Refine activity personalization logic based on real data
- Improve logbook structure and shift handover flow

Phase 3 & 4

Expand & Scale

- Expand activity library with new evidence-based options
- Integrate tracking and outcome measurement tools
- Scale across care settings; conduct formal evaluation
- Publish findings to support sector-wide adoption



The slide features a dark blue background with a teal border on the left and bottom. There are three squares: a large teal square in the top right, a medium dark blue square in the bottom left, and a smaller teal square nested inside the top right square.

Prototype Demo

See how Pepper works in action